CRD

- Treatments – assigned completely at random
- Each experimental unit has same chance of receiving any treatment
- Any difference among experimental units receiving the same treatment – considered **experimental error**
- Appropriate for homogeneous experimental units – lab experiments
  - Environmental condition – easily controlled as compared to field

CRD

- 30 pots (1-30)
- 3 treatments (A-C)
- Each pot – going to receive a single treatment
- Treatments – assigned at random

A	**B**	**B**	**B**	**C**
C	**C**	**A**	**C**	**A**
A	**A**	**B**	**C**	**B**
A	**B**	**C**	**A**	**C**
B	**A**	**A**	**B**	**B**
C	**B**	**C**	**C**	**A**

CRD

- Sources of variation
  - Treatments – source of variation of interest
  - Experimental error – factors beyond control

- $\text{Variability}_{\text{Total}} = \text{Variability}_{\text{Treatment}} + \text{Variability}_{\text{ExperimentalError}}$

- $\text{Variability}_{\text{To}} = \text{Variability}_{t} + \text{Variability}_{e}$

CRD

- Example
- Effects of fertilizer dose on the fresh weight of plants
- Experimental design – CRD
- Independent variable – fertilizer doses
- Treatments – different doses of fertilizer – A, B and C
- Dependent (response) variable – fresh weight of plants

CRD

- Fresh weight (g) of plants under different doses of fertilizer (treatments A-C)

A 32	B 23	B 48	B 17	C 42
C 31	C 41	A 30	C 31	A 39
A 32	A 32	B 29	C 19	B 34
A 46	B 17	C 27	A 41	C 41
B 47	A 33	A 43	B 37	B 25
C 33	B 39	C 32	C 27	A 32

CRD

- Fresh weight (g) of plants under different doses of fertilizer (treatments A-C)
- One-way ANOVA

	A	B	C
1	32	47	31
2	32	23	33
3	46	17	41
4	32	39	27
5	33	48	32
6	30	29	31
7	43	17	19
8	41	37	27
9	39	34	42
10	32	25	41

CRD

Step 1 – make a table of descriptive statistics

Treatment A (Sample 1)	Treatment B (Sample 2)	Treatment C (Sample 3)	Total
n = 10	n = 10	n = 10	$n_{To} = 30$
$\overline{x}_1 = 36$	$\overline{x}_2 = 31.6$	$\overline{x}_3 = 32.4$	
$s^2 = 32.44$	$s^2 = 127.38$	$s^2 = 53.6$	
$\sum x = 360$	$\sum x = 316$	$\sum x = 324$	$\sum x_{To} = 1000$
$(\sum x)^2 = 129600$	$(\sum x)^2 = 99856$	$(\sum x)^2 = 104976$	
$\sum x^2 = 13252$	$\sum x^2 = 11132$	$\sum x^2 = 10980$	$\sum x_{To}{}^2 = 35364$

CRD

- **Step 2** – test for the homogeneity of variance
- By F_{max} test
- F_{max} = largest variance / smallest variance
- $F_{max} = 127.38 / 32.44 \quad = \; 3.93$
- Consult F_{max} table
- a = number of samples / treatments
- v = df of every sample or smallest sample

CRD

- $F_{max} = 3.93$

- Tabled value = 5.34 at $a = 3$ and $v = 9$

- Variances are homogeneous – proceed with ANOVA ✓

CRD

- **Step 3** – calculate CT
- $CT = (\sum x_{To})^2 / n_T = (1000)^2/30 = 33333.33$

- **Step 4** – calculate total SS_T
- $SS_{To} = \sum x_{To}{}^2 - CT \quad = 35364 - 33333.33$

  $= \mathbf{2030.67}$

CRD

- **Step 5** – calculate SS_t

- $SS_t = [(\sum x_1)^2 / n_1] + [(\sum x_2)^2 / n_2] + [(\sum x_3)^2 / n_3] - CT$

- $SS_t = [129600/10] + [99856/10] + [104976/10] - 33333.33$
- $SS_t = 12960 + 9985.6 + 10497.6 - 33333.33$
- $SS_t = 33443.2 - 33333.33$

- $SS_t =$ **109.87**

CRD

- **Step 6** – calculate SS_e
- Individual SS
- $SS_1 = \sum x_1{}^2 - [(\sum x_1)^2 / n_1] = 13252 - [129600/10] = 292$
- $SS_2 = \sum x_2{}^2 - [(\sum x_2)^2 / n_2] = 11132 - [99856/10] = 1146.4$
- $SS_3 = \sum x_3{}^2 - [(\sum x_3)^2 / n_3] = 10980 - [104976/10] = 482.4$

- $SS_e = 292 + 1146.4 + 482.4 =$ **1920.8**
- **Step 7** – cross check value of SS_e
- $SS_{To} = SS_t + SS_e$
- $SS_e = SS_{To} - SS_t = 2030.67 - 109.87 =$ **1920.8**

CRD

- **Step 8** – determine number of degrees of freedom for each SS

- df for $SS_{To} = n_{To} - 1 = 30 - 1 =$ **29**

- df for $SS_t = a - 1 = 3 - 1 =$ **2**

- df for $SS_e = n_{To} - a = 30 - 3 =$ **27**

CRD

- **Step 9** – estimate MS

- $MS_t = SS_t/df_t = 109.87/2$
- $MS_t =$ **54.935**

- $MS_e = SS_e/df_e = 1920.8/27$
- $MS_e =$ **71.14**

CRD

- **Step 10** – compute F

- $F = MS_t/MS_e$
- $F = 54.935/71.14 =$ **0.772**

CRD

- Compare F value (0.772) with critical F value
- Critical F value = 3.35 at p 0.05

CRD

- **Step 11** – construct ANOVA summary table

Source of variation	SS	df	MS	F	P
Treatment	109.87	2	54.935	0.772	>0.05
Error	1920.8	27	71.14		
Total	2030.67	29			

- Compare F value (0.772) with critical F value
- Critical F value = 3.35 at p 0.05
- **Result**: There is no effect of different fertilizer doses on the fresh weight of plants ($F_{2,27} = 0.772$, $p > 0.05$).